

$p \wedge q = p \text{ and } q$
 $p \vee q = p \text{ or } q$
 $p \oplus q = p \text{ XOR } q$
 $p \rightarrow q = \text{if } p \text{ then } q$
 $p \leftrightarrow q = p \text{ if and only if } q$
 $\neg p = \text{not } p$

Prime numbers vs composite numbers n
 has a prime divisor $\leq \sqrt{n}$

Coprime / relatively prime
 if $\gcd = 1$

ORIGINAL NEGATION
 $\forall x, P(x)$ $\exists x, \neg P(x)$
 $\exists x, Q(x)$ $\forall x, \neg Q(x)$

$p \wedge T \equiv p$
 $p \vee F \equiv p$ Identity laws
 $p \vee T \equiv T$
 $p \wedge F \equiv F$ Domination laws

$p \vee p \equiv p$
 $p \wedge p \equiv p$ Idempotent laws
 $\neg(\neg p) \equiv p$ Double negation law

$p \vee q \equiv q \vee p$
 $p \wedge q \equiv q \wedge p$ Commutative laws

$(p \vee q) \vee r \equiv p \vee (q \vee r)$ Associative
 $\wedge \wedge \wedge$

$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ Distributive
 $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

$\neg(p \wedge q) \equiv \neg p \vee \neg q$
 $\neg(p \vee q) \equiv \neg p \wedge \neg q$ De Morgan's Law

$p \vee (p \wedge q) \equiv p$
 $p \wedge (p \vee q) \equiv p$ Absorption laws

$p \vee \neg p \equiv T$
 $p \wedge \neg p \equiv F$ Negation laws

$$\frac{ax \equiv b \pmod{m}}{m \mid ax - b}$$

$a \mid b \Rightarrow a \text{ divides } b$
 $\Rightarrow b \text{ is divisible by } a$
 $\Rightarrow b = k \cdot a \text{ for any } k$

Euclidean Algorithm : $\gcd(a, b) \times \text{lcm}(a, b) = ab$

Extended : If $a \equiv x \pmod{b}$, then $\gcd(a, b) = \gcd(b, x)$

$$\begin{array}{ll} \gcd(133, 49) & 133 - 2 \cdot 49 = 35 \\ \gcd(49, 35) & 49 - 1 \cdot 35 = 14 \\ \gcd(35, 14) & 35 - 2 \cdot 14 = 7 \\ \gcd(14, 7) & 14 - 2 \cdot 7 = 0 \end{array}$$

Backward substitution
 for finding inverse
 if $\gcd = 1$

Bezout's Theorem

If $a, b \in \mathbb{Z}^+$, there are integers r, s such that

$$\gcd(a, b) = ra + sb \quad (=1, \text{ coprime})$$

Fermat's Little Theorem

$$a^{\phi(b)} \equiv 1 \pmod{b}$$

$$a^p \equiv a \pmod{p} \quad \text{any int}$$

$$a^{p-1} \equiv 1 \pmod{p} \quad \text{if } \gcd(a, p) = 1$$

If $x \equiv y \pmod{m}$ and $a \equiv b \pmod{m}$, then $x+a \equiv y+b \pmod{m}$

If $ax \equiv ay \pmod{m}$ and $\gcd(a, m) = 1$, then $x \equiv y \pmod{m}$

Fundamental Theorem of Arithmetic

Every integer greater than 1 can be uniquely represented by a product of prime numbers.

If $a \mid b, b \mid c \Rightarrow$ then $a \mid c$ (transitivity)

If $a \mid b, a \mid c \Rightarrow$ then $a \mid mb + nc$ for any m, n (linearity)

If $a \mid bc$, then $a \mid b$ and $a \mid c$.

$$A \setminus B = A - B$$

$$= \{x \in A \mid x \notin B\}$$

Proof Set Equiv

$$A = B \Leftrightarrow (A \subseteq B) \text{ and } (B \subseteq A)$$

Show $A \subseteq B$

Take $x \in A$. Show $x \in B$.
vice versa

LET $x \in [\text{SET}]$

Proof bijectivity

Find inverse.

$$y = 2x + 3$$

$$x = \frac{y-3}{2}$$

$$f^{-1}(y) = \frac{y-3}{2}$$

$$f(f^{-1}(x)) = x$$

$$f^{-1}(f(x)) = x$$

Proof injectivity.

Start with assumption. Suppose $f(a) = f(b)$

Manipulate until $a = b$ or $a \neq b$

$f(x) = 2x + 3$ on $\mathbb{R}$

Suppose $f(a) = f(b)$ ($a, b \in \mathbb{R}$),

Then $2a + 3 = 2b + 3 \Rightarrow a = b$ ✓

Proof Surjectivity

Start with arbitrary y .

Solve for x .

$f(x) = 2x + 3$ from $\mathbb{R}$ to $\mathbb{R}$.

Let $y \in \mathbb{R}$. $2x + 3 = y$

$$x = \frac{y-3}{2}$$

$x \in \mathbb{R}$ always exists ✓

$\in$ is an element of

$\subseteq$ is a subset of

$$S \subseteq A \rightarrow S \subseteq A \text{ and } S \subseteq B \rightarrow S \subseteq A \cap B$$

$$S \subseteq A \rightarrow \text{If } A \subseteq B \text{ and } B \subseteq A, A = B$$

every element in S is an element of A

$$\text{Power Set } P(\{0, 1, 2\}) = \{\emptyset, \{0\}, \{1\},$$

2^n elements

set of all subsets

$$\{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}$$

$$\text{Cartesian Product } A = \{1, 2\} \times B = \{a, b, c\}$$

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$$

One-to-one $\rightarrow$ injective

Onto $\rightarrow$ surjective

BOTH $\rightarrow$ bijective

$$f: X \rightarrow Y$$

if $f(x_1) = f(x_2)$, then x_1 must $= x_2$

every $y \in Y$ has $x \in X$ where $f(x) = y$

Countable infinities [list]

$\forall$ bijection $f: \mathbb{N} \rightarrow S$

Uncountable infinities [Cantor]

$\forall$ No bijection with $\mathbb{N}$

24 Spring

2) Suppose $a, b \in \mathbb{Z}^+$ with $\gcd(a, b) = 1$.
Let $r \in \mathbb{R}$ st. r^a and r^b are rational numbers. Prove r is rational.

$$r \equiv r' = (r^a)^x (r^b)^y$$

therefore rational
↑
rational
↑
there rational

3) Let S be a set and let $f: S \rightarrow P(S)$.
Prove that $\{s \in S \mid s \notin f(s)\}$ is not in the image of f .

$$P(S) = \{A \mid A \subseteq S\}$$

Power set definition

$$f: S \rightarrow P(S)$$

For each $t \in S$, $f(t) \subseteq S$
 $P(t)$

$$\text{Let set } A = \{s \in S \mid s \notin f(s)\}$$

Then $A \subseteq S$ subset definition
 $A \in P(S)$ subset to power set

Proof $A \notin \text{Im}(f)$:

$$\text{Im}(f) = \{f(t) \mid t \in S\}$$

By def of A , t is element of A iff t is not an element of $f(t)$. If t is in A , t must not be in $f(t)$, vice versa, so, $f(t) \neq A$.

4) Let a be an integer congruent to 1 (mod 3).
Show that a is congruent (mod 9) to one of the 3 numbers: 1, 4, 7. Show that a^3 is also congruent to 1 (mod 9).

$$a \equiv 1 \pmod{3}$$

$$3 \mid a-1$$

$$a-1 = 3n \text{ for any } n$$

$$a = 3n+1$$

$$= 3(3q+r)+1$$

division algorithm
Every integer can be written as $n = 3q+r$ where q is integer and r is 0, 1 or 2

$$= 1 + 9q + 3r$$

$$\equiv 1 + 3r \pmod{9}$$

$$1 + 3(0) = 1$$

$$1 + 3(1) = 4$$

$$1 + 3(2) = 7$$

(mod 9)

$$a^3: 1^3 = 1 \equiv 1 \pmod{9}$$

$$4^3 = 64 \equiv 1 \pmod{9}$$

$$7^3 = 7^2 \cdot 7 \equiv 4 \cdot 7 = 28 \equiv 1 \pmod{9}$$

Proving subset inclusion $A \subseteq B$ (A is smaller)
(B has all of A)

Show $\forall x (x \in A \Rightarrow x \in B)$

Assume $x \in A$, unpack $x = ?$ (by def of A)

Manipulate until $x \in B$.

24 Fall

2) Prove/disprove if sets A, B , then $P(A) \cap P(B) = P(A \cap B)$ where $P(A) = \{S : S \subseteq A\}$

Test $\subseteq$ both ways

$$\rightarrow P(A) \cap P(B) \subseteq P(A \cap B)$$

Assume $S \in P(A) \cap P(B)$.

This means $S \in P(A)$ and $S \in P(B)$.

(S is an element of power set)

By def, $S \subseteq A$ and $S \subseteq B$

(S must be a subset of $A \cap B$)

For every element $x \in S$, $x \in A$ and $x \in B$ (by def).

So $x \in A \cap B$ by def of intersection.

$$\forall x \in S, x \in A \cap B$$

$$\rightarrow P(A \cap B) \subseteq P(A) \cap P(B)$$

(reverse)

4) Prove/disprove: $a, m > 1$, $\gcd(a, m) = 1$,
 $f: \{0, 1, \dots, m-1\} \rightarrow \{0, 1, \dots, m-1\}$
defined by $f(x) = (ax+3) \pmod{m}$ is a bijection.

ONTO: Prove for every $y \in \{0, 1, \dots, m-1\}$ there exists $x \in \{0, 1, \dots, m-1\}$ such that $f(x) = y$

$$ax+3 \equiv y \pmod{m}$$

$$ax \equiv y-3 \pmod{m}$$

Since $\gcd = 1$, there exists $a^{-1} \pmod{m}$.

$$x = a^{-1}(y-3) \pmod{m}$$

$$f(x) = ax+3$$

$$= a(a^{-1}(y-3)) + 3$$

$$= y-3+3$$

$$= y$$

ONE-TO-ONE

Suppose $x_1, x_2 \in \{0, 1, \dots, m-1\}$ satisfy $f(x_1) = f(x_2)$.

$$ax_1+3 \equiv ax_2+3 \pmod{m}$$

$$ax_1 \equiv ax_2 \pmod{m}$$

Properties of Congruence

$$aa^{-1}x_1 \equiv aa^{-1}x_2 \pmod{m}$$

$$x_1 \equiv x_2$$

When $g(x) = (ax^2+3) \pmod{m}$, prove not bijection

ONE-TO-ONE

Let $x_1, x_2 \in \{0, 1, \dots, m-1\}$

and $x_2 = -x_1 (= m-x_1)$

$$g(x_1) = ax_1^2+3 \pmod{m}$$

$$g(x_2) = a(-x_1)^2+3 \pmod{m}$$

$$= ax_1^2+3 \pmod{m}$$

Since $x_1 \neq x_2$ but $g(x_1) = g(x_2)$, $g(x)$ is not one to one.

$\rightarrow$ On the other hand, assume f is injective. Consider $b \in B$. Since $f(b) \in C$ and $f \circ g$ is surjective, there exists $a \in A$ such that $f(g(a)) = f \circ g(a) = f(b)$. Since f is injective, it implies $g(a) = b$, thus g is surjective.

22 Fall

2) Prove/disprove: if $f: B \rightarrow C$, $g: A \rightarrow B$

a)

and $f \circ g$ is one to one, then g must be one-to-one.

$$A \xrightarrow{g} B \xrightarrow{f} C$$

(Assume injective)

Suppose $x_1, x_2 \in A$ that satisfy $g(x_1) = g(x_2)$ (def of 1-1)

$$f \circ g(x_1) = f(g(x_1)) = f(g(x_2)) = f \circ g(x_2)$$

Since $f \circ g$ is injective, by definition if $f \circ g(x_1) = f \circ g(x_2)$, then $x_1 = x_2$.

5b) Prove $a^8 + b^8 + c^8 = 555554$ has no solution. (Hint: mod 5)

$$a^8 + b^8 + c^8 = (a^4)^2 + (b^4)^2 + (c^4)^2$$

$$x^2 \pmod{5}$$

$$x \equiv 0 \pmod{5} \text{ if } 5 \mid x \text{ or } x^4 \equiv 1 \pmod{5} \text{ by FLT}$$

$$x^8 \equiv 0 \text{ or } x^8 \equiv 1$$

$$a^8 + b^8 + c^8 \equiv 0, 1, 2, 3 \pmod{5}$$

$$55554 \equiv 4 \pmod{5}$$

2b) $\therefore$ then f must be one-to-one

$$A \xrightarrow{g} B \xrightarrow{f} C$$

$$1 \rightarrow 1 \rightarrow 1$$

one to one
one to one
one to one

25 Spring

3a) If $f \circ g$ is surjective, then f must also be surjective.

Consider $c \in C$. Since $f \circ g$ is onto, we know there is $a \in A$ such that $f \circ g(a) = c$. From this, we know there exists $g(a) \in B$ such that $f(g(a)) = c$. $\checkmark$

b) If $f \circ g$ is a bijection, then g is surjective iff f is injective

$\rightarrow$ Assume g is surjective.

Let $b_1, b_2 \in B$ st $f(b_1) = f(b_2)$.

Since g is surjective, we know there exists $a_1, a_2 \in A$

such that $b_1 = g(a_1)$ and $b_2 = g(a_2)$. Thus $f \circ g(a_1) = f \circ g(a_2)$.

Thus $b_1 = g(a_1) = g(a_2) = b_2$, f is injective.